

Phase 2 – SIEM Dashboard Analysis using Splunk on Kali

In this phase, we used Splunk on Kali Linux to analyze logs collected from the victim machine (Metasploitable3).

After multiple attempts to install and configure the Splunk Universal Forwarder on the victim, we encountered persistent compatibility errors due to the outdated operating system and unsupported architecture.

As a result, we decided to adopt a manual approach: log files were securely transferred using scp from the victim to the attacker machine (Kali), and then uploaded to Splunk for visualization and analysis.

Step 1: Install Splunk on Kali Linux

We downloaded and installed Splunk Enterprise on Kali Linux using the .deb package. After accepting the license and starting the service, we accessed the Splunk Web Interface at <http://localhost:8000>.

```
(walah@kali)-[~]
$ sudo dpkg -i splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb
[sudo] password for walah:
Selecting previously unselected package splunk.
(Reading database ... 442134 files and directories currently installed.)
Preparing to unpack splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb ...
Unpacking splunk (9.3.2) ...
Setting up splunk (9.3.2) ...
complete
(walah@kali)-[~]
$
```

```
walah@kali: ~
File Actions Edit View Help
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'privKeySecure.pem'
-----
Signature ok
subject=/CN=kali/O=SplunkUser
Getting CA Private Key
writing RSA key
PYTHONHTTPVERIFY is set to 0 in splunk-launch.conf disabling certificate val
idation for the httplib and urllib libraries shipped with the embedded Python
interpreter; must be set to "1" for increased security
Done

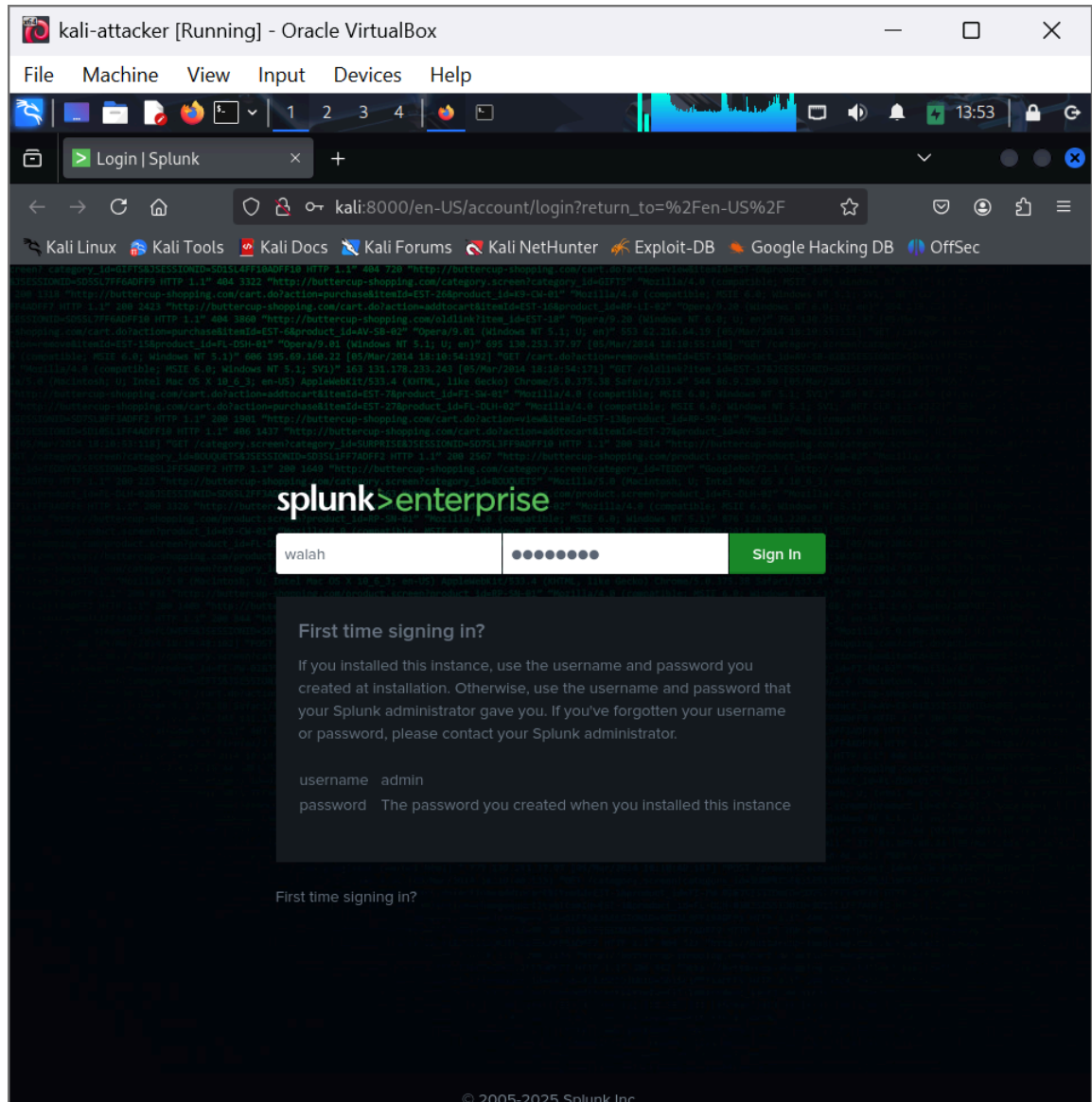
Waiting for web server at http://127.0.0.1:8000 to be available.....
Done

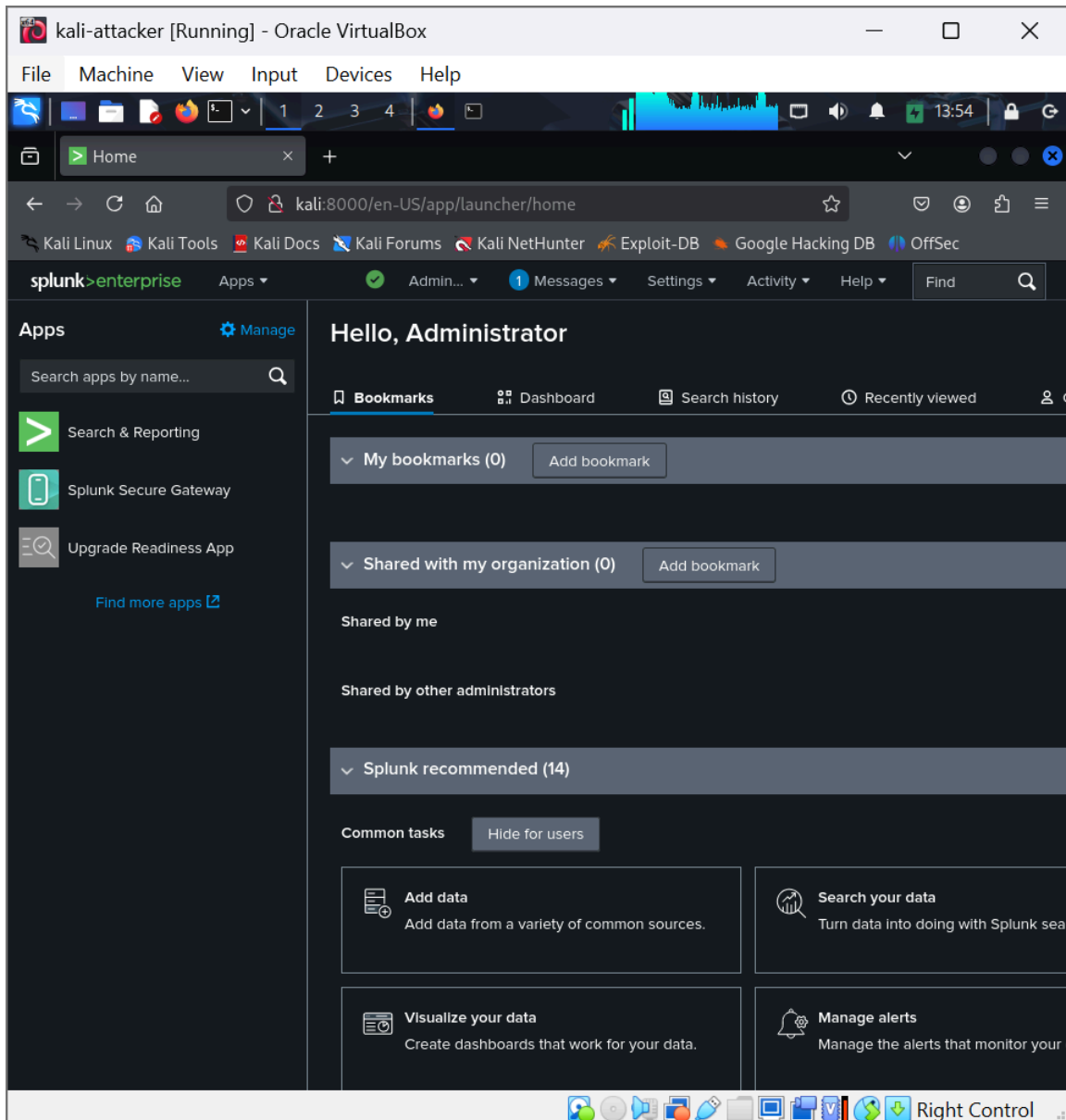
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://kali:8000
(walah@kali)-[~]
$
```

Step 2: Access Splunk Web Interface

We logged into Splunk Web Interface using the admin credentials set during the initial setup.

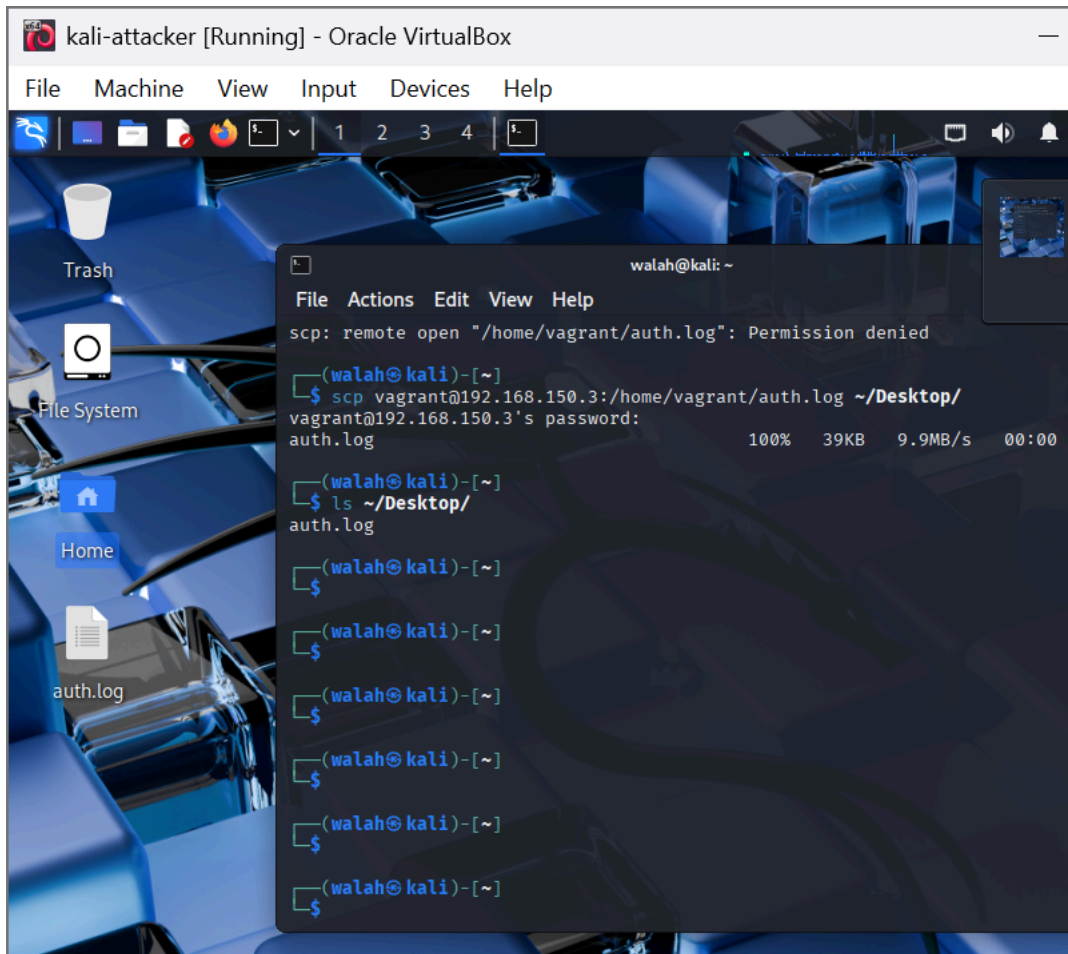




Step 3: Manually Uploading Log Files to Splunk

We used the `scp` command on Kali to copy the file `/home/vagrant/auth.log` from the victim (Metasploitable3) to our attacker machine.

Then, we used Splunk Web's `Add Data > Upload File` feature to ingest this file.



kali-attacker [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

1234

20:45

Problem loading pageAdd Data - Select Source

kali:8000/en-US/manager/search/adddatamethods/selectsource?in

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splunk>enterpriseAppsAdmin...MessagesSettingsActivityHelpFind

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< BackNext >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: **auth.log**

Select File

Drop your data file here

The maximum file upload size is 500 Mb

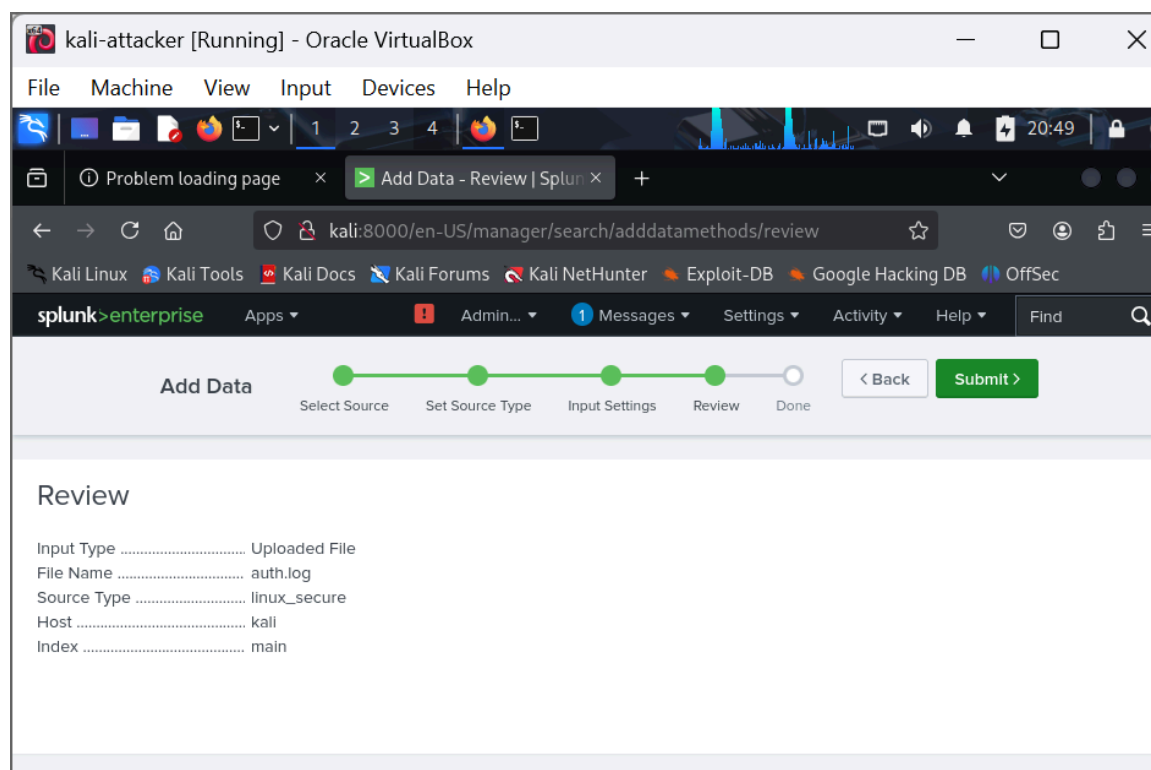
File Successfully Uploaded

FAQ

> What kinds of files can the Splunk platform index?

> What is a source?

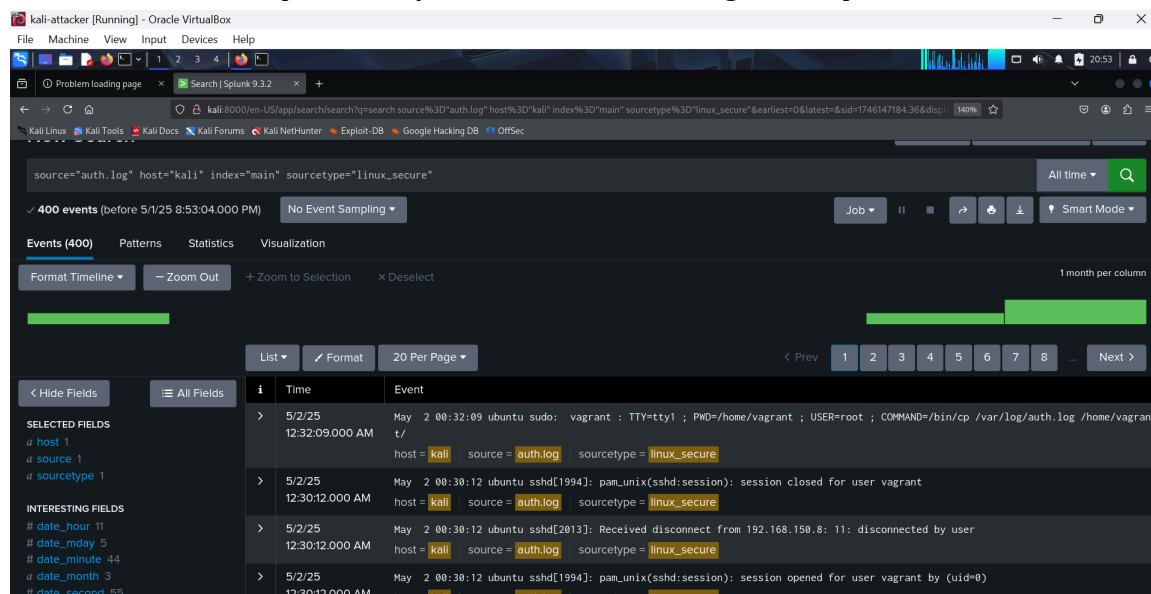
> How do I get remote data onto my Splunk platform instance?



The log file was uploaded successfully to Splunk and is ready for searching and analysis.

Step 4: Search and Analyze Logs

We used the Splunk Search & Reporting app to analyze logins, failed authentications, and SSH behavior. This helped identify successful and failed login attempts from the attack.



kali-attacker [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Text Editor | Simple Text Editor

Search | Splunk 9.3.2

kali:8000/en-US/app/search?q=search source%3D"auth.log" host%3D"kali" index%3D"main" sourcetype%3D"linux_secure"&earliest=0&latest=&sid=1746147184.36&disp= 140%

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< Hide Fields All Fields List Format 20 Per Page

	i	Time	Event
# linecount 1			
# pid 100+			
a process 12			
a punct 71			
a splunk_server 1			
# timeendpos 1			
# timestartpos 1			
# uid 1			
14 more fields			
+ Extract New Fields			
		5/2/25 12:29:30.000 AM	May 2 00:29:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root by vagrant(uid=0)
		5/2/25 12:29:30.000 AM	May 2 00:29:30 ubuntu sudo: vagrant : TTY=ttty1 ; PwD=/home/vagrant ; USER=root ; COMMAND=/bin/cp /var/log/auth.log /home/vagrant/
		5/2/25 12:25:14.000 AM	May 2 00:25:14 ubuntu sshd[1931]: pam_unix(sshd:session): session closed for user vagrant
		5/2/25 12:25:14.000 AM	May 2 00:25:14 ubuntu sshd[1949]: Received disconnect from 192.168.150.8: 11: disconnected by user
		5/2/25 12:25:14.000 AM	May 2 00:25:14 ubuntu sshd[1931]: pam_unix(sshd:session): session opened for user vagrant by (uid=0)
		5/2/25 12:25:14.000 AM	May 2 00:25:14 ubuntu sshd[1931]: Accepted password for vagrant from 192.168.150.8 port 34680 ssh2
		5/2/25 12:23:02.000 AM	May 2 00:23:02 ubuntu login[1766]: pam_unix(login:session): session opened for user vagrant by LOGIN(uid=0)
		5/2/25 12:22:51.000 AM	May 2 00:22:51 ubuntu login[1766]: FAILED LOGIN (1) on '/dev/tty1' FOR 'UNKNOWN', Authentication failure

Step 5: Dashboard Visualization

This spike in the graph indicates a significant increase in events related to unauthorized access attempts. The rise reflects the attacker's repeated login attempts on the SSH service, which were logged by the system.

